Hazards, Exposure, Vulnerability and Disaster Risk

My reviewer for Disaster Readiness and Risk Reduction (DRRR). For us in the Philippines, this is not just for the exam.

Hazard vs disaster

A hazard is anything that could harm people, property or the environment. It is a possible danger, not yet damage, like a typhoon far out in the Pacific.

A disaster happens when a hazard actually hits a community and the damage is so serious that the community cannot cope using only its own resources. If that same typhoon floods a whole town, destroys houses and the barangay needs outside help, that is now a disaster.

The risk idea

Disaster risk is the chance that a hazard will lead to losses:

- Risk = Hazard x Exposure x Vulnerability / Capacity
- Hazard: the dangerous event itself, like ground shaking or storm surge.
- Exposure: who and what are located in the danger area. A school built beside a riverbank is more exposed to flooding than one on higher ground.
- Vulnerability: how easily those exposed people or things get damaged. A house made of light materials, a family with no savings, or an elderly person living alone are more vulnerable.
- Capacity: the strengths and resources a community has to prepare, respond and recover, like trained rescuers, early warning systems, evacuation centers and good roads.

Capacity is in the denominator, so raising capacity lowers risk. We cannot stop a typhoon, but we can lower exposure and vulnerability and build capacity.

Natural vs human-induced hazards

- Natural hazards come from natural processes. Examples: earthquakes, volcanic eruptions, typhoons, tsunamis, landslides triggered by heavy rain.
- Human-induced hazards come from human activity or negligence. Examples: fires caused by faulty electrical wiring, oil spills, chemical leaks from factories, and structural collapse from poor construction.

Who watches the hazards

- PAGASA (Philippine Atmospheric, Geophysical and Astronomical Services Administration) handles weather forecasts, typhoon tracking and rainfall warnings.
- PHIVOLCS (Philippine Institute of Volcanology and Seismology) monitors earthquakes, volcanoes and tsunamis. It records quakes, issues tsunami advisories, and raises or lowers volcano alert levels. Both agencies are under the Department of Science and Technology (DOST).

PAGASA tropical cyclone wind signals

The Tropical Cyclone Wind Signal (TCWS) shows the wind strength expected in an area. Higher number, stronger wind, less lead time.

- Signal No. 1: strong winds of 39 to 61 km/h expected within 36 hours. Minor damage, like twigs and light objects flying.
- Signal No. 2: gale-force winds of 62 to 88 km/h expected within 24 hours. Light roofing may be blown off.
- Signal No. 3: storm-force winds of 89 to 117 km/h expected within 18 hours. Moderate to heavy damage.
- Signal No. 4: typhoon-force winds of 118 to 184 km/h expected within 12 hours. Severe damage.
- Signal No. 5: winds of 185 km/h or higher expected within 12 hours. Extreme, possibly catastrophic damage.

Earthquake preparedness

- Before: secure tall cabinets to the wall, know the safe spots in each room, join school earthquake drills, and agree on a family meeting place.
- During: do Drop, Cover and Hold On. Drop to the floor, take cover under a sturdy table, and hold on until the shaking stops. Stay away from glass windows. If outside, move to an open area away from posts and buildings.
- After: expect aftershocks, check yourself and others for injuries, avoid damaged buildings, and if you live near the coast and the shaking was strong, move to higher ground right away because a tsunami may follow.

Typhoon preparedness

- Before: monitor PAGASA updates, trim tree branches near the house, charge phones and power banks, store clean water and food, and follow early evacuation orders from the LGU.
- During: stay indoors and away from windows, unplug appliances if flooding starts, and never cross flooded roads or rivers.
- After: wait for the all-clear, stay away from fallen power lines, boil drinking water if the supply is unsure, and watch for diseases like leptospirosis when wading in floodwater.

Go-bag contents

A go-bag is a ready emergency bag you grab when you must leave fast. It should last about three days:

- Bottled water and ready-to-eat food like canned goods with easy-open lids and biscuits
- Flashlight, extra batteries and a whistle for signaling
- First aid kit and personal medicines
- Copies of important documents in a waterproof plastic pouch
- Extra clothes, a blanket and a raincoat
- Power bank, some cash in small bills, and a list of emergency contact numbers
- Hygiene items like soap, alcohol, face masks and sanitary pads

Quick check

- Q: What is the difference between a hazard and a disaster? A: A hazard is a potential source of harm; a disaster is when a hazard seriously damages a community beyond its ability to cope alone.

- Q: In the risk formula, what happens to risk when capacity increases? A: Risk decreases, because capacity is the divisor.
- Q: What wind speed range goes with Signal No. 3? A: 89 to 117 km/h, expected within 18 hours.
- Q: Which agency monitors earthquakes and volcanoes? A: PHIVOLCS, under DOST.
- Q: What should you do during an earthquake? A: Drop, Cover and Hold On, away from windows and heavy objects.